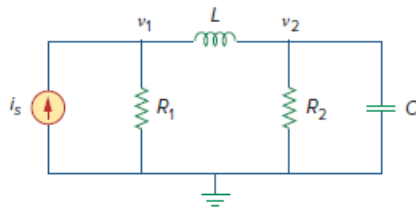


Problem

Using Fig. 16.85, design a problem to help other students better understand solving for node voltages by working in the s -domain.

Figure 16.85:



Step-by-step solution

Step 1 of 1

Using Fig. 16.56, design a problem to help other students better understand solving for node voltages by working in the s -domain.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

Find the node voltages v_1 and v_2 in the circuit of Fig. 16.56 using Laplace transform technique. Assume that $i_s = 12e^{-t}u(t)$ A and that all initial conditions are zero.

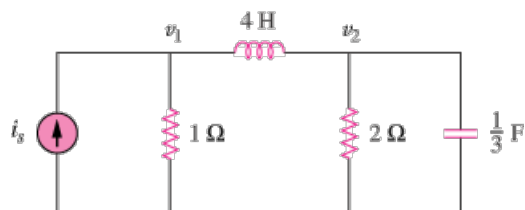
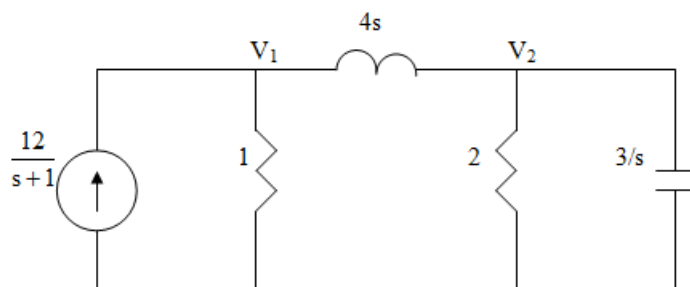


Figure 16.56

Solution

The s -domain version of the circuit is shown below.



At node 1,

$$\frac{12}{s+1} = \frac{V_1}{1} + \frac{V_1 - V_2}{4s} \longrightarrow \frac{12}{s+1} = V_1 \left(1 + \frac{1}{4s} \right) - \frac{V_2}{4s} \quad (1)$$

At node 2,

$$\frac{V_1 - V_2}{4s} = \frac{V_2}{2} + \frac{s}{3} V_2 \longrightarrow V_1 = V_2 \left(\frac{4}{3}s^2 + 2s + 1 \right) \quad (2)$$

Substituting (2) into (1),

$$\frac{12}{s+1} = V_2 \left[\left(\frac{4}{3}s^2 + 2s + 1 \right) \left(1 + \frac{1}{4s} \right) - \frac{1}{4s} \right] = \left(\frac{4}{3}s^2 + \frac{7}{3}s + \frac{3}{2} \right) V_2$$

$$V_2 = \frac{9}{(s+1)(s^2 + \frac{7}{3}s + \frac{9}{2})} = \frac{A}{(s+1)} + \frac{Bs+C}{(s^2 + \frac{7}{3}s + \frac{9}{2})}$$

$$9 = A(s^2 + \frac{7}{4}s + \frac{9}{8}) + B(s^2 + s) + C(s + 1)$$

Equating coefficients:

$$s^2 : \quad 0 = A + B$$

$$s : \quad 0 = \frac{7}{4}A + B + C = \frac{3}{4}A + C \quad \longrightarrow \quad C = -\frac{3}{4}A$$

$$\text{constant} : \quad 9 = \frac{9}{8}A + C = \frac{3}{8}A \quad \longrightarrow \quad A = 24, B = -24, C = -18$$

$$V_2 = \frac{24}{(s+1)} - \frac{24s+18}{(s^2 + \frac{7}{4}s + \frac{9}{8})} = \frac{24}{(s+1)} - \frac{24(s + 7/8)}{(s + \frac{7}{8})^2 + \frac{23}{64}} + \frac{3}{(s + \frac{7}{8})^2 + \frac{23}{64}}$$

Taking the inverse of this produces:

$$v_2(t) = \underline{[24e^{-t} - 24e^{-0.875t} \cos(0.5995t) + 5.004e^{-0.875t} \sin(0.5995t)]u(t)V}$$

Similarly,

$$V_1 = \frac{9(\frac{4}{3}s^2 + 2s + 1)}{(s+1)(s^2 + \frac{7}{4}s + \frac{9}{8})} = \frac{D}{(s+1)} + \frac{Es + F}{(s^2 + \frac{7}{4}s + \frac{9}{8})}$$

$$9(\frac{4}{3}s^2 + 2s + 1) = D(s^2 + \frac{7}{4}s + \frac{9}{8}) + E(s^2 + s) + F(s + 1)$$

Equating coefficients:

$$s^2 : \quad 12 = D + E$$

$$s : \quad 18 = \frac{7}{4}D + E + F \text{ or } 6 = \frac{3}{4}D + F \quad \longrightarrow \quad F = 6 - \frac{3}{4}D$$

$$\text{constant} : \quad 9 = \frac{9}{8}D + F \text{ or } 3 = \frac{3}{8}D \quad \longrightarrow \quad D = 8, E = 4, F = 0$$

$$V_1 = \frac{8}{(s+1)} + \frac{4s}{(s^2 + \frac{7}{4}s + \frac{9}{8})} = \frac{8}{(s+1)} + \frac{4(s + 7/8)}{(s + \frac{7}{8})^2 + \frac{23}{64}} - \frac{7/2}{(s + \frac{7}{8})^2 + \frac{23}{64}}$$

Thus,

$$v_1(t) = \underline{[8e^{-t} + 4e^{-0.875t} \cos(0.5995t) - 5.838e^{-0.875t} \sin(0.5995t)]u(t)V}$$